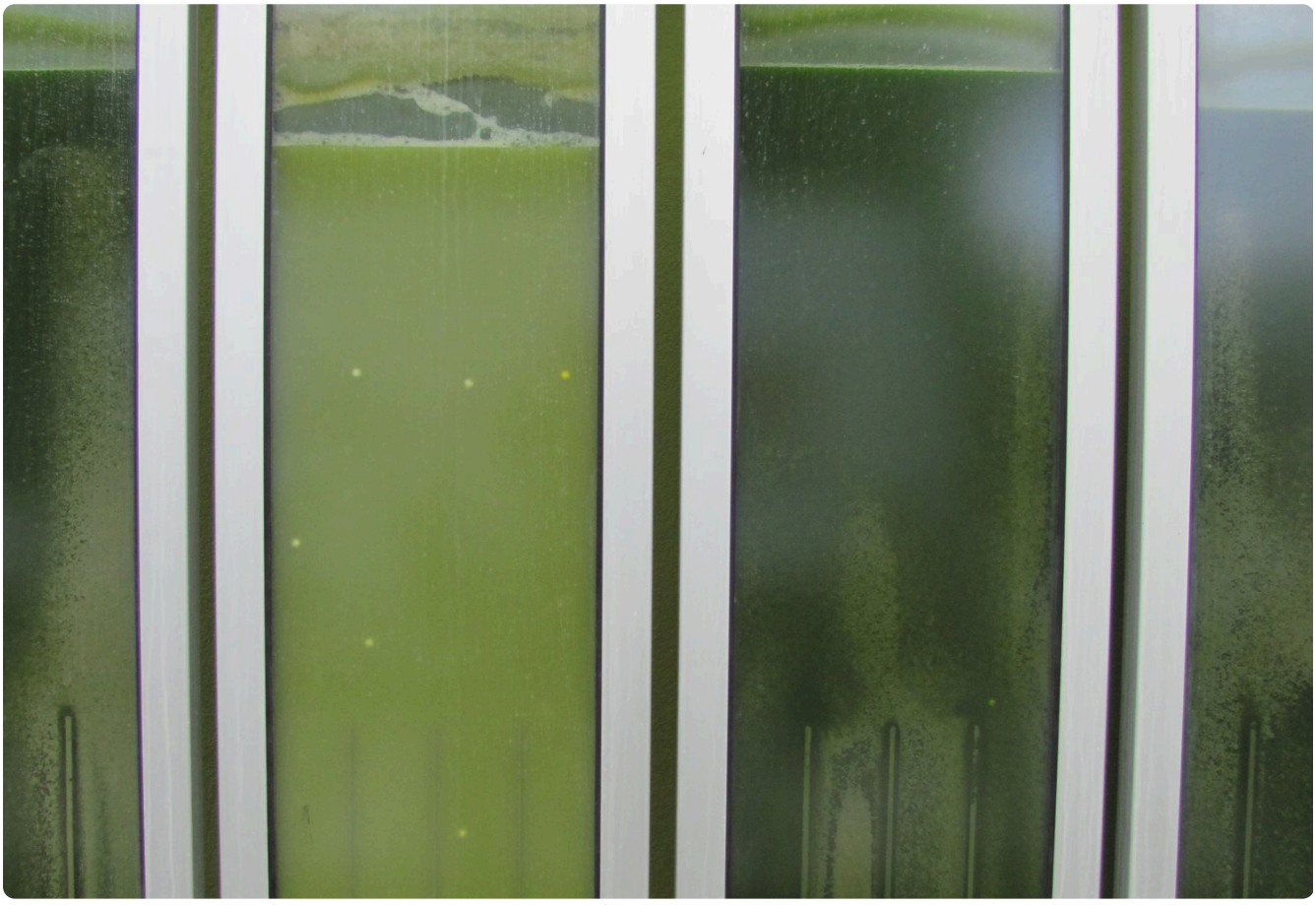
DESIGN COMPUTED · FREE GUIDE

Algae facades: a wall that grows its own heat and harvest

The free guide to the reel on the BIQ house in Hamburg: how a glass wall full of algae shades a room, makes heat and biomass, where it has been built, and a cheap experiment you can run on a window sill.



How it works

A bioreactive facade replaces ordinary glazing with flat glass tanks full of water and microalgae. Sunlight goes in, the algae use part of it to grow (photosynthesis), and the rest turns into heat in the water.

That gives the wall three jobs at once. It shades the rooms behind it, because the green water absorbs light. It collects heat, which a heat exchanger carries to a store or to the building's heating. And the algae grow into biomass, which can be harvested.

Pumps move water and nutrients through the panels, and CO2 is added so the algae keep growing. The tanks are placed on the sunny sides, and the shading changes as the algae grow denser.

The real numbers (BIQ house, Hamburg, 2013)

The BIQ house was built for the International Building Exhibition (IBA) in Hamburg. It has 129 flat glass bioreactors, each about 2.5 m by 0.7 m, on its two sunny sides, about 200 m² in total (Arup, Colt International, SSC Ltd).

Monitoring found that about 21 % of the sunlight became heat and about 4 % became biomass, roughly half of what was modelled at design stage. The system covered about a third of the heating needs of the building's 15 homes. It is a pilot: promising, not magic.

Built examples

BIQ house, Hamburg (2013)	Splitterwerk architects with Arup and Colt: the first full-scale bioreactive facade, 129 panels on two faces.
Urban Algae Folly (ecoLogicStudio)	ecoLogicStudio: an algae-filled ETFE pavilion; the cushions darken as the algae grow.
Photo.Synth.Etica, Dublin Castle (2018)	ecoLogicStudio: a curtain of 16 bioplastic photobioreactor modules on the Printworks building.
Algae research facades	Several university groups test panel size, flow and shading control; start with the papers in Go further.

Try it yourself

1. Find a clear glass jar or a flat food box and a window that gets direct sun for several hours.
2. Buy a small amount of Chlorella or spirulina culture from a school-science or aquarium supplier, or use pond water kept in daylight.
3. Fill the container with water, add a little liquid plant fertiliser and the algae, and put it on the sill.
4. Stir gently every day and add a few drops of fertiliser each week.
5. Every day photograph the jar against a white sheet and note how the colour deepens.
6. Put a thermometer in the jar and another in a jar of clear water beside it. Compare the temperatures at noon.
7. After a week, sketch how you would put 100 of these in a wall: where do the pipes go and how do you clean them?

The rule to remember

When a wall bakes in the sun, don't only hide from the heat. Ask what could live on it.

Treating it as decoration	The panels need pumps, cleaning and monitoring: plan the maintenance on day one.
Putting it on a shaded side	The system lives on sunlight; use the sunniest faces.
Expecting big energy output	Real results were about half of the model: size your expectations on measured data.

Go further

World's first bioreactive facade debuts in Hamburg, Arup
https://www.arup.com/news-and-events/world-first-bioreactive-facade-debuts-in-hamburg

BIQ house, Archello
https://archello.com/project/biq-house

Algae-powered architecture, Ingenia
https://ingenia.org.uk/articles/algae-powered-architecture/

Photo.Synth.Etica, ecoLogicStudio

https://aasarchitecture.com/2019/02/photo-synth-etica-by-ecologicstudio

Credits
Photos: see the reel's caption (Wikimedia Commons, free licences).
Facts: Arup, Archello, Ingenia, Designcurial.

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